

UVM Phasing and Synchronization

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Discussion on UVM Verbosity

- Using run time option +UVM_VERBOSITY=UVM_NONE
- UVM_NONE, UVM_LOW, UVM_MEDIUM (Default)
- UVM_HIGH, UVM_FULL, UVM_DEBUG
- <https://www.edaplayground.com/x/dCwe>

Common Phases in UVM

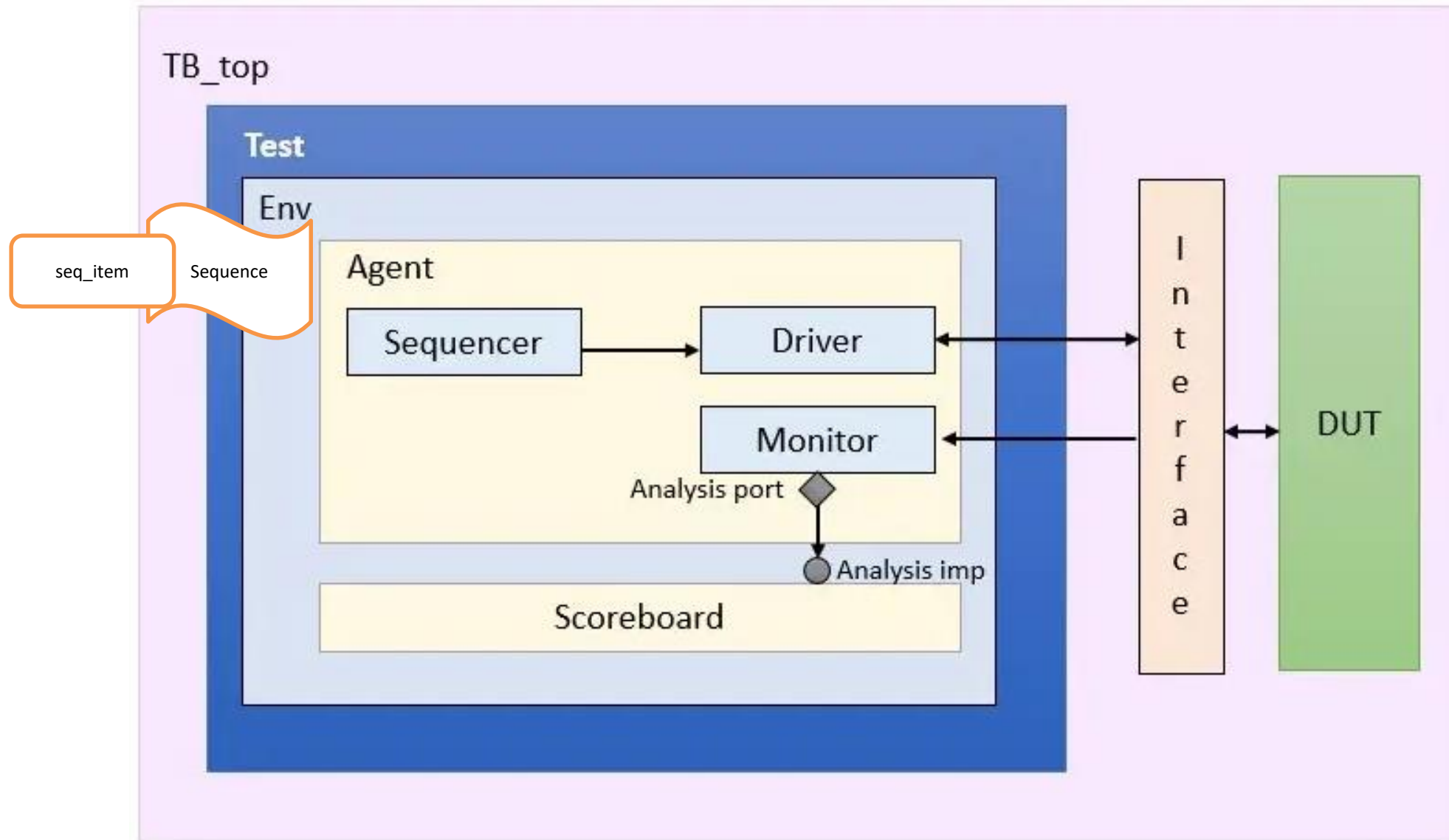
- Common Phases in UVM are
 - uvm_build_phase
 - uvm_connect_phase
 - uvm_end_of_elaboration_phase
 - uvm_start_of_simulation_phase
 - uvm_run_phase
 - uvm_extract_phase
 - uvm_check_phase
 - uvm_report_phase
 - uvm_final_phase

UVM Common Phases

The common phases are the set of function and task phases that all `uvm_components` execute together.

uvm_build_phase	Create and configure of testbench structure
uvm_connect_phase	Establish cross-component connections.
uvm_end_of_elaboration_phase	Fine-tune the testbench.
uvm_start_of_simulation_phase	Get ready for DUT to be simulated.
uvm_run_phase	Stimulate the DUT.
uvm_extract_phase	Extract data from different points of the verification environment.
uvm_check_phase	Check for any unexpected conditions in the verification environment.
uvm_report_phase	Report results of the test.
uvm_final_phase	Tie up loose ends.

Recall UVM based Testbench



UVM_Build_Phase

- **UVM_Build_Phase is executed in Top-Down fashion.**
- **The functions of UVM_Build_Phase are**
 - Instantiate sub-components, instantiate register model.
 - Get configuration values for the component being built.
 - Set configuration values for sub-components.

UVM_Connect_Phase

- **UVM_Connect_Phase is executed in Bottom-Up fashion**
- **UVM_Connect_Phase is used to**
 - Connect TLM ports and exports.
 - Connect TLM initiator sockets and target sockets.
 - Connect register model to adapter components.

UVM_Start_of_Simulation_Phase

- UVM_Start_of_Simulation_Phase is executed in Bottom_Up fashion
- UVM_Start_of_Simulation_Phase is used to
 - Display environment topology
 - Set debugger breakpoint
 - Set initial run-time configuration values.

UVM_Run_Phase

- UVM_Run_Phase is executed in Bottom_Up fashion
- UVM_Run_Phase is used to
 - Implement component's behavior that is exhibited for the entire run-time, across the various run-time phases
- **The run phase terminates in one of two ways.**
 - All run_phase objections are dropped
 - Timeout

Lab Exercise

Lab 3 Understanding UVM Phases

To understand and implement `build_phase`, `connect_phase`, and `run_phase` in UVM.